

6 (a) $kx_o = Mg$

$$F = ma$$

$$k(x_o + x) - Mg = -Ma$$

$$k(x_o + x) - Mg = -Ma$$

$$Mg + kx - Mg = -Ma$$

$$a = -\frac{k}{M}x$$

(b) Acceleration is directly proportional to displacement as k/M is a constant

Acceleration is always in the opposite direction to displacement as indicated by the negative sign

(c) $\omega^2 = \frac{k}{M}$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{M}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{150}{2}}$$

$$= 1.38 \text{ Hz}$$

(d) (i) 5.0 Hz

(ii) Lightly damped forced oscillation

Maximum amplitude at the natural frequency of the spring mass system, must label $f = 1.38\text{Hz}$

(iii) Translate vertically upwards